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On Analytical Solutions of Matrix Riccati Equations

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Abstract—Matrix Riccati equations appear in numerous applications, especially in control engineering. In this paper we derive analytical formulas for exact solutions of algebraic and differential matrix Riccati equations. These solutions are expressed in terms of matrix transfer functions of appropriate linear dynamical systems.

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1. INTRODUCTION

One of the main results of this paper is an analytical formula for the exact matrix solution $R \in \mathbb{R}^{n\times n}$ of the following *algebraic* matrix Riccati equation:

$$L + A^{\mathrm{T}}R + RA - RBM^{-1}B^{\mathrm{T}}R = 0 \tag{1.1}$$

where

$$A \in \mathbb{R}^{n\times n}, \qquad B \in \mathbb{R}^{n\times m}, \qquad L \in \mathbb{R}^{n\times n}, \qquad M \in \mathbb{R}^{m\times m}$$

and the matrices L and M are positive definite ($L = L^{\mathrm{T}}$, $M = M^{\mathrm{T}}$, $L > 0$, $M > 0$).

This quadratic matrix equation is closely related to the problem of feedback stabilization (see [1, 2, 11]) of the following linear control system:

$$\dot{x} = Ax + Bu \tag{1.2}$$

where $x \in \mathbb{R}^n$ is a state vector and $u \in \mathbb{R}^m$ is a control.

Namely, the stabilization problem is to find a matrix $K \in \mathbb{R}^{m\times n}$ such that the feedback

$$u = Kx \tag{1.3}$$

is stabilizing, which means that the closed-loop system

$$\dot{x} = (A + BK)x$$

is asymptotically stable.

In the early 1960s Kalman in the USA [8] and Letov in the Soviet Union [12] suggested a new approach to finding a stabilizing feedback for the linear system (1.2) based on the use of optimal control state-space methods. Namely, they demonstrated that the optimal control feedback minimizing the quadratic functional

$$J(x,u) := \int_0^{+\infty} \frac{1}{2}\big(\langle x, Lx\rangle + \langle u, Mu\rangle\big)\,dt \tag{1.4}$$

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is also a stabilizing feedback. Moreover, this stabilizing optimal feedback can be expressed in terms of the solution R of the matrix Riccati equation (1.1):

$$u = -M^{-1}B^{\mathrm{T}}Rx. \tag{1.5}$$

This approach of "analytic design of regulators" (the term coined by Letov) helped to overcome limitations of the classical "frequency-domain" methods for stabilization of multivariable control systems.

In this paper we use the classical "frequency-domain" concepts of matrix transfer functions

$$G_0(s) := (sI - A)^{-1}, \qquad G_u(s) := G_0(s)B = (sI - A)^{-1}B \tag{1.6}$$

in order to obtain a solution R of the algebraic matrix Riccati equation (1.1) (here I denotes the identity matrix). It seems that in many engineering applications such transfer functions can be effectively evaluated without the need to identify the matrix A.

Of course, some results relating "frequency-domain" methods and linear–quadratic optimal control problems were obtained long ago (see, for example, the work by Willis and Brockett [16]). However, we have not seen in the existing literature an analytical formula for the solution of the Riccati equation in terms of matrix transfer functions.

Let us define the matrix function

$$G(i\omega) := \bigl(I + G_u(i\omega)M^{-1}G_u^{\mathrm{H}}(i\omega)L\bigr)^{-1} \tag{1.7}$$

where $i^2 = -1$, $\omega \in \mathbb{R}$, and G^{H} is the Hermitian conjugate of G.

It is well known that for the asymptotically controllable linear system (1.2) there exists a positive definite matrix R that is a solution of the matrix equation (1.1) (see, for example, [2, 9]).

Theorem 1.1. *Let system* (1.2) *be asymptotically controllable. Then the matrix*

$$R = \left(\int_{-\infty}^{+\infty} G(i\omega)G_u(i\omega)M^{-1}G_u^{\mathrm{H}}(i\omega)\,d\omega\right)^{-1}\left(\pi I - \int_{-\infty}^{+\infty} G(i\omega)G_0(i\omega)\,d\omega\right) \tag{1.8}$$

is a solution of the algebraic matrix Riccati equation (1.1).

We prove this formula in Section 3 by using classical elementary methods of the Laplace transform, including an inversion formula for it. However, we also need a relation between an adjoint variable in the Pontryagin maximum principle and a gradient of the value function of an infinite-horizon optimal control problem. For the completeness of exposition we provide a short derivation of such relations in Section 2.

In Section 4 we consider a differential matrix Riccati equation

$$\dot{R} = -L - A^{\mathrm{T}}R - RA + RBM^{-1}B^{\mathrm{T}}R \tag{1.9}$$

and derive an analytic formula for a solution of this differential equation with the quadratic right-hand side.

Finally, in Section 5 we demonstrate another approach to the derivation of analytical solutions of the algebraic matrix Riccati equation. This method also provides a representation of the optimal stabilizing feedback (1.3) entirely in terms of matrix transfer functions (1.6) without the need to identify the matrix B (see (1.5)).

We use the traditional notation A^{T} for the transpose of a matrix A and $A^{\mathrm{H}} = \overline{A^{\mathrm{T}}}$ for the Hermitian transpose of a complex matrix A. The identity matrix is denoted by I, the inner product of vectors $x, y \in \mathbb{R}^n$ is denoted by $\langle x, y\rangle$, and the Euclidean norm $\|x\| = \langle x, x\rangle^{1/2}$.

2. MAXIMUM PRINCIPLE AND VALUE FUNCTION

One of the important mathematical discoveries of the 20th century took place at the Steklov Mathematical Institute when Pontryagin and his collaborators Boltyanskii and Gamkrelidze discovered and proved the maximum principle as a necessary condition for a general optimal control problem [13]. The development of optimal control theory influenced many mathematical fields. It also provided new mathematical tools for classical control theory by introducing space–time models of control systems along with classical "frequency-domain" ones.

Now we introduce two standard optimal control problems and describe the relation between the adjoint variable in the maximum principle and the gradients of value functions.

Consider a control system given by the differential equation

$$\dot{x}(t) = f(x(t), u(t)) \tag{2.1}$$

where $x \in \mathbb{R}^n$ is a state vector, $u(t) \in \mathbb{U}$ is control, and $\mathbb{U}$ is a subset of $\mathbb{R}^m$. We impose the traditional continuity assumptions on $f(x,u)$ and $f_x(x,u)$ and the growth condition which provides a boundedness of the trajectory $x(t)$ for any admissible control $u(t)$.

Finite-horizon optimal control problem. Minimize the following functional over the set of all admissible controls and trajectories starting from x_0, $x(0) = x_0$:

$$J^\theta(x,u) := \ell(x(\theta)) + \int_0^\theta L(x(t), u(t))\, dt \tag{2.2}$$

where $\theta > 0$ is fixed.

Infinite-horizon optimal control problem. Minimize the following functional over the set of all admissible controls and trajectories starting from x_0, $x(0) = x_0$:

$$J^\infty(x,u) = \int_0^{+\infty} L(x(t), u(t))\, dt. \tag{2.3}$$

We also consider a value function for the finite-horizon problem

$$V^\theta(t,x) := \inf\left\{ \ell(x(\theta)) + \int_t^\theta L(x(\tau), u(\tau))\, d\tau : \ x(t) = x, \ \forall\, u(\cdot) \right\}, \tag{2.4}$$

where the infimum is taken over all admissible controls and trajectories satisfying the initial condition $x(t) = x$, and a value function for the infinite-horizon problem

$$V^\infty(x) := \inf\left\{ \int_0^{+\infty} L(x(\tau), u(\tau))\, d\tau : \ x(0) = x, \ \forall\, u(\cdot) \right\}. \tag{2.5}$$

Assume that $(x_0(t), u_0(t))$ is an optimal trajectory/control pair in problem (2.2) or (2.3). In order to write necessary optimality conditions for this optimal pair, we define a Pontryagin Hamiltonian (or pseudo-Hamiltonian in Clarke's terminology [3, 4])

$$H(p,x,u) := \langle p, f(x,u)\rangle + L(x,u).$$

It is convenient to write these conditions in the form of a minimum (instead of maximum) principle. Namely, for the optimal pair $(x_0(t), u_0(t))$ in the finite-horizon problem (2.2) there exists an adjoint variable $p(t)$ that satisfies the adjoint equation

$$\dot{p}(t) = -H_x(p(t), x_0(t), u_0(t)), \tag{2.6}$$

the boundary condition

$$p(\theta) = \ell_x(x_0(\theta)), \tag{2.7}$$

and is such that the following minimum principle holds for a.a. $t \in [0, \theta]$:

$$H(p(t), x_0(t), u_0(t)) = \min_{u \in U} H(p(t), x_0(t), u). \tag{2.8}$$

If the value function $V^\theta(t, x)$ is differentiable at $x_0(t)$, then the adjoint variable satisfies the equality

$$p(t) = V_x^\theta(t, x_0(t)). \tag{2.9}$$

It is well known that in general value functions are not differentiable. Then equality (2.9) should be replaced by an inclusion with the set-valued Clarke generalized gradient [6, 4] instead of the gradient V_x (see also [10] for related results).

In this section we provide a proof of equality (2.9) for finite- and infinite-horizon problems; it will be used in the following sections. This proof also demonstrates the usefulness of the *sliding variation* method in optimal control [7]. A slight modification of this method and a multidirectional mean-value inequality [5] from nonsmooth analysis can be used to obtain a generalization of (2.9) for nonsmooth value functions in terms of Clarke generalized gradients.

Theorem 2.1. *Let for an optimal pair $(x_0(t), u_0(t))$ in the finite-horizon problem the partial derivative $V_x^\theta(t, x_0(t))$ exist for any $t \in [0, \theta]$. Then there exists an adjoint variable $p(t)$ satisfying* (2.6), (2.7), *and* (2.9) *such that the optimal control satisfies the minimum principle* (2.8).

Proof. Choose an arbitrary admissible control $u(t)$, a vector $e \in \mathbb{R}^n$, and $\lambda \in [0, 1]$ and consider solutions $x_\lambda(t)$ and z_λ of the following differential equations:

$$\dot{x_\lambda}(t) = (1-\lambda) f(x_\lambda(t), u_0(t)) + \lambda f(x_\lambda(t), u(t)), \qquad x_\lambda(0) = x_0 + \lambda e, \tag{2.10}$$

$$\dot{z_\lambda}(t) = (1-\lambda) L(x_\lambda(t), u_0(t)) + \lambda L(x_\lambda(t), u(t)), \qquad z_\lambda(0) = 0. \tag{2.11}$$

Note that the variation $\delta x(t) := \frac{\partial x_\lambda(t)}{\partial \lambda}$ is a solution of the linear differential equation

$$\dot{\delta x}(t) = A(t)\delta x(t) + \Delta f(t), \qquad \delta x(0) = e, \tag{2.12}$$

where

$$A(t) := f_x(x_0(t), u_0(t)), \qquad \Delta f(t) = f(x_0(t), u(t)) - f(x_0(t), u_0(t)).$$

It is easy to show the following approximation result: For any $\varepsilon > 0$ there exists an admissible control $u^\varepsilon(t)$ such that the corresponding trajectories $x^\varepsilon(t)$ and $z^\varepsilon(t)$ of

$$\dot{x}^\varepsilon(t) = f(x^\varepsilon(t), u^\varepsilon(t)), \quad x^\varepsilon(0) = x_0 + \lambda e, \qquad \dot{z}^\varepsilon(t) = L(x^\varepsilon(t), u^\varepsilon(t)), \quad z^\varepsilon(0) = 0,$$

approximate x_λ and z_λ:

$$\max_{t \in [0,\theta]} \big(\|x^\varepsilon(t) - x_\lambda(t)\| + |z^\varepsilon(t) - z_\lambda(t)| \big) < \varepsilon.$$

Now we fix $t \in (0, \theta)$. It follows from the optimality of $x_0(t)$ that

$$V^\theta(x_0) = V^\theta(t, x_0(t)) + \int_0^t L(x_0(\tau), u_0(\tau))\, d\tau$$

and from the approximation result that

$$V^\theta(x_0 + \lambda e) \le V^\theta(x_\lambda(t)) + \int_0^t \big((1-\lambda)L(x_\lambda(\tau), u_0(\tau)) + \lambda L(x_\lambda(\tau), u(\tau))\big)\, d\tau.$$

Subtracting the previous equality from this inequality, dividing by λ, and taking the limit as $\lambda \downarrow 0$, after a rearrangement of terms we obtain

$$\langle V_x^\theta(t, x_0(t)), \delta x(t)\rangle - \langle V^\theta(0, x_0), e\rangle + \int_0^t \big(\langle L_x(x_0(\tau), u_0(\tau)), \delta x(\tau)\rangle + \Delta L(\tau)\big)\, d\tau \ge 0 \tag{2.13}$$

for any vector e and any admissible control u; here

$$\Delta L(\tau) = L(x_0(\tau), u(\tau)) - L(x_0(\tau), u_0(\tau)).$$

We replace δx by its representation in terms of the fundamental matrix $\Phi(t, s)$ of the linear equation (2.12), the function Δf, and the vector e. Then we can rewrite inequality (2.13) as follows:

$$\langle p(0) - V_x^\theta(x_0), e\rangle + \int_0^t \big(H(p(\tau), x_0(\tau), u(\tau)) - H(p(\tau), x_0(\tau), u_0(\tau))\big)\, d\tau \ge 0 \tag{2.14}$$

where the variable $p(\tau)$ is defined as

$$p(\tau) := \Phi^{\mathrm{T}}(t, \tau) V_x^\theta(t, x_0(t)) + \int_\tau^t \Phi^{\mathrm{T}}(\sigma, \tau) L_x(x_0(\sigma), u_0(\sigma))\, d\sigma. \tag{2.15}$$

It is clear that p is a solution of the adjoint equation (2.6) and satisfies

$$p(0) = V_x^\theta(0, x_0) \tag{2.16}$$

and (2.9). Note that because of the initial condition (2.16) the adjoint variable does not depend on t, which implies due to (2.9) that p also satisfies the boundary condition (2.7) since $V^\theta(\theta, x) = \ell(x)$.

It follows from (2.14) for $e = 0$ that for any admissible control u

$$\int_0^t \big(H(p(\tau), x_0(\tau), u(\tau)) - H(p(\tau), x_0(\tau), u_0(\tau))\big)\, d\tau \ge 0,$$

which implies that u_0 satisfies the minimum principle (2.8). $\square$

Note that if the pair $(x_0(t), u_0(t))$ is optimal for the infinite-horizon problem (2.3), then for any $\theta > 0$ this pair is also optimal for the finite-horizon problem with the functional

$$V^\infty(x(\theta)) + \int_0^\theta L(x(t), u(t))\, dt,$$

with $V^\theta(0, x) = V^\infty(x)$.

Applying Theorem 2.1 to this optimal control problem, we conclude that there exists an adjoint variable $p(t)$ satisfying the adjoint equation (2.6), the minimum principle (2.8), the boundary condition (2.7) with $\ell(x) = V^\infty(x)$, and equality (2.16), which becomes

$$p(0) = V_x^\infty(x_0). \tag{2.17}$$

It follows from this initial condition that the adjoint variable p does not depend upon θ. Thus, from Theorem 2.1 we deduce the following

Corollary 2.1. *Let $(x_0(t), u_0(t))$ be an optimal pair for the infinite-horizon optimal control problem of minimization* (2.3). *Then there exists an adjoint variable $p(t)$ satisfying the adjoint equation* (2.6), *the minimum principle* (2.8), *and the equality*

$$p(t) = V_x^\infty(x_0(t)), \qquad t \geq 0. \tag{2.18}$$

3. ALGEBRAIC MATRIX RICCATI EQUATION

In this section we prove the analytical formula (1.8) for the solution of the algebraic matrix Riccati equation (1.1). Recall that this equation is related to the following optimal control problem.

Linear–quadratic optimal control problem. It is well known [1, 2, 9] that for the optimal control problem (1.2)–(1.4) and asymptotically controllable system the value function is quadratic in x:

$$V^\infty(x) = \frac{1}{2}\langle x, Rx\rangle \tag{3.1}$$

where a symmetric positive definite matrix R is a solution of the algebraic matrix Riccati equation (1.1).

Then it follows from the minimum principle (Corollary 2.1) that for the optimal control pair $(x(t), u(t))$

$$u(t) = -M^{-1}B^{\mathrm{T}}p(t), \qquad p(t) = Rx(t). \tag{3.2}$$

Thus the optimal trajectory $x(t)$ and adjoint variable $p(t)$ satisfy the adjoint system

$$\begin{aligned} \dot{x} &= Ax - BM^{-1}B^{\mathrm{T}}p, \\ \dot{p} &= -A^{\mathrm{T}}p - Lx. \end{aligned} \tag{3.3}$$

We recall the Laplace transform of a function $f(t)$ and its derivative,

$$\widehat{f}(s) = \int\limits_0^{+\infty} f(t)e^{-st}\,dt, \qquad \widehat{\dot{f}}(s) = s\widehat{f}(s) - f(0),$$

and a general inverse theorem for the Laplace transform in terms of the Bromwich integral (see [15]),

$$\frac{1}{2\pi i}\int\limits_{c-i\omega}^{c+i\omega} \widehat{f}(s)e^{st}\,ds = \frac{f(t-0)+f(t+0)}{2}.$$

We also need a special case of this formula for $t = 0$ in the case of a continuous function $f(t)$ that is absolutely integrable on $[0, \pi)$ and vanishes for $t < 0$:

$$\frac{1}{2\pi}\int\limits_{-\infty}^{+\infty} \widehat{f}(i\omega)\,d\omega = \frac{f(0)}{2}. \tag{3.4}$$

Note that these inverse formulas are manifestations of the results on the inverse Fourier transform.

Proof of Theorem 1.1. For a solution (x,p) of system (3.3) we write equations for their Laplace transforms

$$s\widehat{x}(s) - x_0 = A\widehat{x}(s) - BM^{-1}B^{\mathrm{T}}\widehat{p}(s),$$

$$s\widehat{p}(s) - p_0 = -A^{\mathrm{T}}\widehat{p}(s) - L\widehat{x}(s)$$

where $x_0 = x(0)$ and $p_0 = p(0)$.

This system can be rewritten as

$$\widehat{x}(s) = G_0(s)x_0 - G_u(s)M^{-1}B^{\mathrm{T}}\widehat{p}(s),$$

$$\widehat{p}(s) = G_1(s)p_0 - G_1(s)L\widehat{x}(s)$$

where G_0 and G_u were defined in (1.6) and

$$G_1(s) := (sI + A^{\mathrm{T}})^{-1}.$$

We solve this system with respect to the unknown $\widehat{x}$ and $\widehat{p}$ to find

$$\bigl(I - G_u(s)M^{-1}B^{\mathrm{T}}G_1(s)L\bigr)\widehat{x}(s) = G_0(s)x_0 - G_u(s)M^{-1}B^{\mathrm{T}}G_1(s)p_0.$$

We replace here s by $i\omega$ and use the fact that

$$G_1(i\omega) = -G_0(i\omega)^{\mathrm{H}}$$

to obtain

$$\widehat{x}(i\omega) = G(i\omega)G_0(i\omega)x_0 + G(i\omega)G_u(i\omega)M^{-1}G_u^{\mathrm{H}}(i\omega)p_0 \tag{3.5}$$

where G is defined in (1.7).

We integrate (3.5) in ω and due to (3.4) obtain

$$\pi x_0 = \int_{-\infty}^{+\infty} \widehat{x}(i\omega)\,d\omega = \left(\int_{-\infty}^{+\infty} G(i\omega)G_0(i\omega)\,d\omega\right)x_0 + \left(\int_{-\infty}^{+\infty} G(i\omega)G_u(i\omega)M^{-1}G_u^{\mathrm{H}}(i\omega)\,d\omega\right)p_0. \tag{3.6}$$

Note that the convergence of the integral L_1 defined below follows from the convergence of the integral of the function $\widehat{x}(i\omega)$ in (3.5) and the obvious convergence of the integral L_2:

$$L_1 := \int_{-\infty}^{+\infty} G(i\omega)G_0(i\omega)\,d\omega, \qquad L_2 := \int_{-\infty}^{+\infty} G(i\omega)G_u(i\omega)M^{-1}G_u^{\mathrm{H}}(i\omega)\,d\omega. \tag{3.7}$$

Then we can rewrite (3.6) in the form

$$p(0) = L_2^{-1}(\pi I - L_1)x(0).$$

But using relation (3.2) between p and x, we obtain

$$R = L_2^{-1}(\pi I - L_1),$$

which is exactly formula (1.8). $\square$

Solution formalism for a general algebraic matrix Riccati equation. It is obvious that the same method can be applied to a more general linear system

$$\begin{aligned} \dot{x} &= Ax + Bp, \\ \dot{p} &= Cx + Dp \end{aligned} \tag{3.8}$$

where $x \in \mathbb{R}^n$, $p \in \mathbb{R}^m$, and A, B, C, and D are matrices of appropriate dimensions.

We try to find a matrix $R \in \mathbb{R}^{m\times n}$ such that the linear subspace of solutions of (3.8) satisfying

$$p(t) = Rx(t) \tag{3.9}$$

is nonempty for any initial condition $x(0)$.

We leave aside the question of existence of such a matrix R assuming its existence. In this sense we provide some formal computations assuming the convergence of all integrals involved and the invertibility of appropriate matrices.

Substituting (3.9) into the second equation in (3.8), we obtain

$$R(Ax(t) + BRx(t)) = Cx(t) + DRx(t).$$

Since we assumed that this is true for any initial condition $x(0)$, we see that the matrix R should satisfy the following algebraic matrix Riccati equation:

$$C - RA + DR - RBR = 0. \tag{3.10}$$

Thus, we are aimed at writing a formula for a matrix solution R of this general algebraic matrix Riccati equation.

We write system (3.8) in terms of the Laplace transforms $\widehat{x}$ and $\widehat{p}$ of x and p using the notation $x(0) = x_0$ and $p(0) = p_0$:

$$\begin{aligned} s\widehat{x}(s) - x_0 &= A\widehat{x}(s) + B\widehat{p}(s), \\ s\widehat{p}(s) - p_0 &= C\widehat{x}(s) + D\widehat{p}(s). \end{aligned}$$

Then it can be written in terms of the matrix transfer functions G_0 (1.6) and

$$G_2(s) := (sI - D)^{-1} \tag{3.11}$$

as

$$\begin{aligned} \widehat{x}(s) &= G_0(s)x_0 + G_0(s)B\widehat{p}(s), \\ \widehat{p}(s) &= G_2(s)p_0 + G_2(s)C\widehat{x}(s). \end{aligned} \tag{3.12}$$

From this system of equations we obtain

$$\widehat{x}(s) = H_1(s)G_0(s)x_0 + H_1(s)G_0(s)BG_2(s)p_0 \tag{3.13}$$

where

$$H_1(s) = \bigl(I - G_0(s)BG_2(s)C\bigr)^{-1}. \tag{3.14}$$

Replacing s with $i\omega$ in (3.13) and integrating it from $-\infty$ to $+\infty$, due to (3.4) we obtain

$$\pi x_0 = \left(\int\limits_{-\infty}^{+\infty} H_1(i\omega)G_0(i\omega)\,d\omega\right)x_0 + \left(\int\limits_{-\infty}^{+\infty} H_1(i\omega)G_0(i\omega)BG_2(i\omega)\,d\omega\right)p_0. \tag{3.15}$$

We denote

$$L_1 := \int_{-\infty}^{+\infty} H_1(i\omega)G_0(i\omega)\,d\omega, \qquad L_2 := \int_{-\infty}^{+\infty} H_1(i\omega)G_0(i\omega)BG_2(i\omega)\,d\omega. \tag{3.16}$$

Then it follows from (3.9)–(3.15) that

$$L_2R = \pi I - L_1.$$

Assuming that $m \le n$ and L_2 has maximal rank, we can resolve the previous equation to find R and to obtain the following

Theorem 3.1. *Assume that the linear system* (3.8) *has an asymptotically stable solution satisfying* (3.9) *for any initial* $x(0)$, *integrals* (3.16) *converge,* $m \le n$, *and the matrix* L_2 *has rank* m. *Then the solution* R *of the matrix quadratic equation* (3.10) *has the representation*

$$R = (L_2^{\mathrm{T}}L_2)^{-1}L_2^{\mathrm{T}}(\pi I - L_1). \tag{3.17}$$

Note that in the case when $m = n$ and the system of equations (3.8) coincides with the adjoint system (3.3), formula (3.17) coincides with (1.8).

Now we find another representation of the solution R of (3.10) under different assumptions by finding from (3.12) the formula for $\widehat{p}$:

$$\widehat{p}(s) = H_2(s)G_2(s)p_0 + H_2(s)G_2(s)CG_0(s)x_0 \tag{3.18}$$

where

$$H_2(s) := \bigl(I_m - G_2(s)CG_0(s)\bigr)^{-1} \tag{3.19}$$

and I_m is the $m \times m$ identity matrix.

Then, substituting $i\omega$ for s in (3.18) and integrating it from $-\infty$ to $+\infty$, due to (3.4) we obtain

$$\pi p_0 = \left(\int_{-\infty}^{+\infty} H_2(i\omega)G_2(i\omega)\,d\omega\right)p_0 + \left(\int_{-\infty}^{+\infty} H_2(i\omega)G_2(i\omega)CG_0(i\omega)\,d\omega\right)x_0.$$

Recalling that $p_0 = Rx_0$ and x_0 is arbitrary, we obtain

$$\left(\pi I_m - \int_{-\infty}^{+\infty} H_2(i\omega)G_2(i\omega)\,d\omega\right)R = \int_{-\infty}^{+\infty} H_2(i\omega)G_2(i\omega)CG_0(i\omega)\,d\omega. \tag{3.20}$$

Thus, we proved the following

Theorem 3.2. *Assume that the linear system* (3.8) *has an asymptotically stable solution satisfying* (3.9) *for any initial* $x(0)$, *the integrals in* (3.20) *converge, and the matrix multiplying* R *in* (3.20) *is invertible. Then the solution* R *of the matrix quadratic equation* (3.10) *has the representation*

$$R = \left(\pi I_m - \int_{-\infty}^{+\infty} H_2(i\omega)G_2(i\omega)\,d\omega\right)^{-1} \int_{-\infty}^{+\infty} H_2(i\omega)G_2(i\omega)CG_0(i\omega)\,d\omega. \tag{3.21}$$

4. DIFFERENTIAL MATRIX RICCATI EQUATIONS

For matrices

$$A \in \mathbb{R}^{n\times n}, \qquad B \in \mathbb{R}^{n\times m}, \qquad C \in \mathbb{R}^{m\times n}, \qquad D \in \mathbb{R}^{m\times m}$$

we consider a differential matrix Riccati equation [14]

$$\dot{R}(t) = C + DR(t) - R(t)A - R(t)BR(t), \qquad R(0) = R_0, \tag{4.1}$$

where $R(t) \in \mathbb{R}^{m\times n}$ is a matrix-valued function.

In this section we present an analytical formula for a solution of the differential equation (4.1) by relating it to a solution of the linear system (3.8).

We make the following assumption about (4.1):

Assumption 4.1. There exists $\theta > 0$ such that the solution $R(t)$ exists and has the maximal rank on $[0, \theta]$.

It is easy to check that under this assumption for any $x_0 \in \mathbb{R}^n$ there exists a unique solution $(x(t), p(t))$ of (3.8) with the initial conditions

$$x(0) = x_0, \qquad p(0) = p_0 := R_0 x_0 \tag{4.2}$$

satisfying on $[0, \theta]$

$$p(t) = R(t)x(t). \tag{4.3}$$

To derive a formula for $R(t)$, we fix $t \in (0, \theta]$ and consider that the functions x and p are defined on the interval $[0, t]$ as solutions of (3.8) with the initial condition (4.2) but they vanish outside this interval.

Then the Laplace transforms of the derivatives of these functions are

$$\widehat{\dot{x}}(s) = x(t)e^{-st} - x_0 + s\widehat{x}(s), \qquad \widehat{\dot{p}}(s) = p(t)e^{-st} - p_0 + s\widehat{p}(s), \tag{4.4}$$

and from system (3.8) we have

$$(sI - A)\widehat{x}s = x_0 - e^{-st}x(t) + B\widehat{p}(s), \qquad (sI - D)\widehat{p}(s) = p_0 - e^{-st}p(t) + C\widehat{x}(s).$$

Using the transfer functions G_0 (1.6) and G_2 (3.11), we obtain the system

$$\begin{aligned} \widehat{x}(s) &= G_0(s)(x_0 - e^{-st}x(t)) + G_0(s)B\widehat{p}(s), \\ \widehat{p}(s) &= G_2(s)(p_0 - e^{-st}p(t)) + G_2(s)C\widehat{x}(s). \end{aligned}$$

Using the matrix function H_1 (3.14), we can find a solution $(\widehat{x}, \widehat{p})$ of this system, for example,

$$\widehat{x}(s) = H_1(s)G_0(s)(x_0 - e^{-st}x(t)) + H_1(s)G_0(s)BG_2(s)(p_0 - e^{-st}p(t)). \tag{4.5}$$

Recall (see (3.4)) that

$$\int_{-\infty}^{+\infty} \widehat{x}(i\omega)\, d\omega = \pi x_0, \qquad \int_{-\infty}^{+\infty} \widehat{x}(i\omega)e^{i\omega t}\, d\omega = \pi x(t). \tag{4.6}$$

First we substitute $i\omega$ for s in (4.5) and integrate it from $-\infty$ to $+\infty$; then we multiply (4.5) by $e^{i\omega t}$ and integrate to obtain, due to (4.6),

$$\begin{aligned} \pi x_0 &= L_1 x_0 - M_1(t)x(t) + L_2 p_0 - M_2(t)p(t), \\ \pi x(t) &= M_1(-t)x_0 - L_1 x(t) + M_2(-t)p_0 - L_2 p(t) \end{aligned} \tag{4.7}$$

where the matrices L_1 and L_2 are defined by (3.16) and

$$M_1(t) := \int_{-\infty}^{+\infty} H_1(i\omega)G_0(i\omega)e^{-i\omega t}\,d\omega, \qquad M_2(t) := \int_{-\infty}^{+\infty} H_1(i\omega)G_0(i\omega)BG_2(i\omega)e^{-i\omega t}\,d\omega. \tag{4.8}$$

We use (4.2) and (4.3) to find a relation between x_0 and $x(t)$, namely,

$$(\pi I - L_1 - L_2R_0)x_0 = -M_1(t)x(t) - M_2(t)R(t)x(t).$$

We assume that the matrix

$$N(R_0) := (\pi I - L_1 - L_2R_0)^{-1} \tag{4.9}$$

exists. Then from the previous equality and the second equation in (4.7) we have

$$(\pi I + L_1)x(t) = \big(M_1(-t) + M_2(-t)R_0\big)N(R_0)\big(-M_1(t) - M_2(r)R(t)\big)x(t) - L_2R(t)x(t).$$

This implies that

$$Q(t,R_0)R(t) = -\big(\pi I + L_1 + (M_1(-t) + M_2(-t)R_0)N(R_0)M_1(t)\big)$$

where

$$Q(t,R_0) := \big(M_1(-t) + M_2(-t)R_0\big)N(R_0)M_2(t) + L_2. \tag{4.10}$$

Assume that the matrix $Q(t,R_0) \in \mathbb{R}^{n\times m}$ has a full rank m. Then from the previous relations we have

$$R(t) = -\big(Q(t,R_0)^{\mathrm{T}}Q(t,R_0)\big)^{-1}Q(t,R_0)^{\mathrm{T}}\big(\pi I + L_1 + (M_1(-t) + M_2(-t)R_0)N(R_0)M_1(t)\big). \tag{4.11}$$

Thus, we proved the following

Theorem 4.1. *Let Assumption* 4.1 *hold, integrals* (3.16) *and* (4.8) *converge, the matrix* $N(R_0)$ *exist,* $m \le n$, *and* $Q(t,R_0)$ (4.10) *have rank* m *for all* $t \in (0,\theta]$. *Then the solution* $R(t)$ *of the differential matrix Riccati equation* (4.1) *is given by* (4.11).

Note that it follows from formula (4.11) and the residue integration for integrals in (4.8) that the entries of the solution $R(t)$ are rational functions of trigonometric and exponential functions.

Finite-horizon linear–quadratic optimal control problem. We consider a classical optimal control problem of minimization of the functional

$$\frac{1}{2}\langle x(\theta), R_0x(\theta)\rangle + \frac{1}{2}\int_0^\theta \big(\langle x(t), Lx(t)\rangle + \langle u(t), Mu(t)\rangle\big)\,dt$$

over the set of pairs x, u of the control system (1.2). It is assumed that the matrices R_0, L, and M are positive definite.

It is well known [1, 9] that for this problem the value function (2.4) is quadratic in x:

$$V^\theta(t,x) = \frac{1}{2}\langle x, R(t)x\rangle$$

where $R(t)$ satisfies the differential matrix Riccati equation (1.9) and the initial condition

$$R(\theta) = R_0. \tag{4.12}$$

It is obvious that $R(t) = \widetilde{R}(\theta - t)$ where $\widetilde{R}(t)$ satisfies the general differential matrix Riccati equation (4.1) and the initial condition $\widetilde{R}(0) = R_0$, and we should replace B by $BM^{-1}B^{\mathrm{T}}$, D by A^{T}, A by $-A$, and C by $-L$ in (4.1).

Then, in order to use formula (4.11), in the definitions of the matrices L_1, L_2, M_1, and M_2 we need to replace $G_0(s)$, $G_2(s)$, and $H_1(s)$ (3.14) by the following functions, respectively:

$$\widetilde{G}_0(s) = (sI + A)^{-1}, \qquad \widetilde{G}_2(s) = (sI - A^{\mathrm{T}})^{-1}, \qquad \widetilde{H}_1(s) = \bigl(I - \widetilde{G}_0(s)B\widetilde{G}_2(s)C\bigr)^{-1}.$$

Note that

$$\widetilde{G}_0(-i\omega) = -G_0(i\omega), \qquad \widetilde{G}_2(-i\omega) = G_0^{\mathrm{H}}(i\omega), \qquad \widetilde{H}_1(-i\omega) = G(i\omega)$$

where G was defined in (1.7).

Then we can find the matrices L_1 and L_2 (3.16) in terms of the matrix transfer functions G_0 and G_u (1.6):

$$L_1 := -\int_{-\infty}^{+\infty} G(i\omega)G_0(i\omega)\,d\omega, \qquad L_2 := -\int_{-\infty}^{+\infty} G(i\omega)G_u(i\omega)M^{-1}G_u^{\mathrm{H}}(i\omega)\,d\omega. \tag{4.13}$$

Similarly we can write the matrix functions M_1 and M_2 (4.8) in terms of G, G_0, and G_u as follows:

$$M_1(t) := -\int_{-\infty}^{+\infty} G(i\omega)G_0(i\omega)e^{i\omega t}\,d\omega, \qquad M_2(t) := -\int_{-\infty}^{+\infty} G(i\omega)G_u(i\omega)M^{-1}G_0^{\mathrm{H}}(i\omega)e^{i\omega t}\,d\omega. \tag{4.14}$$

Then, assuming that the matrix $Q(t, R_0)$ (4.10) is invertible for all $t \in [0, \theta]$ and using Theorem 4.1, we obtain

Corollary 4.1. *Let integrals* (4.13) *and* (4.14) *converge and the matrices* $Q^{-1}(t, R_0)$ *and* $N(R_0)$ *exist. Then the solution* $R(t)$ *of the differential matrix Riccati equation* (1.9) *with the initial condition* (4.12) *is represented as follows*:

$$R(t) = Q^{-1}(\theta - t, R_0)\bigl(\pi I + L_1 + (M_1(t-\theta) + M_2(t-\theta)R_0)N(R_0)M_1(\theta - t)\bigr). \tag{4.15}$$

5. OPTIMAL FEEDBACK FORMULA IN TERMS OF TRANSFER FUNCTIONS

It is interesting to note that even if the solution (1.8) of the algebraic matrix Riccati equation is expressed in terms of the matrix transfer functions (1.6), the stabilizing feedback (1.5) formally requires the knowledge of the matrix B in (1.2). Of course, this matrix can be evaluated by using the known transfer functions G_0 and G_u, but it is desirable to find an optimal stabilizing feedback in an explicit form in terms of transfer functions.

In this section we demonstrate the derivation of such a formula by using an approach different from the one in the previous sections. This approach is based on the reduction of the original linear–quadratic optimal control problem to a linear–quadratic infinite-dimensional minimization problem in the space of Fourier transforms of controls.

Let $f\colon (-\infty, +\infty) \to \mathbb{R}^n$ be an integrable function; its Fourier transform will be denoted by a capital letter F or by $\mathcal{F}[f]$,

$$\mathcal{F}[f](\omega) = F(\omega) := \int_{-\infty}^{+\infty} f(t)e^{-i\omega t}\,d\omega.$$

We consider a trajectory/control pair x, u in (1.2) with the initial condition $x(0) = x_0$ and assume that $x(t) = 0$ and $u(t) = 0$ for $t < 0$. Then, of course,

$$\mathcal{F}[\dot{x}](\omega) = i\omega\mathcal{F}[x](\omega) - x_0$$

and for the Fourier transforms X and U of x and u we have

$$X(\omega) = G_0(i\omega)x_0 + G_u(i\omega)U(\omega). \tag{5.1}$$

In order to state the main result of this section, we use the following notation:

$$S(i\omega) := \bigl(M + G_u^{\mathrm{H}}(i\omega)LG_u(i\omega)\bigr)^{-1}. \tag{5.2}$$

Note that $S(i\omega)$ is a positive definite matrix.

Theorem 5.1. *Let system* (1.2) *be asymptotically controllable. Then the optimal stabilizing feedback for problem* (1.2)–(1.4) *is represented by the formula*

$$u = -\frac{1}{\pi}\int\limits_{-\infty}^{+\infty} S(i\omega)G_u^{\mathrm{H}}(i\omega)LG_0(i\omega)\,d\omega\,x \tag{5.3}$$

and the solution R of the matrix Riccati equation is represented by the formula

$$R = \frac{1}{2\pi}\int\limits_{-\infty}^{+\infty} \bigl(G_0^{\mathrm{H}}(i\omega)LG_0(i\omega) - G_0^{\mathrm{H}}(i\omega)LG_u(i\omega)S(i\omega)G_u^{\mathrm{H}}(i\omega)LG_0(i\omega)\bigr)\,d\omega. \tag{5.4}$$

Proof. Using Parseval's identity, we can write the functional $J(x,u)$ in (1.4) in terms of the Fourier transforms of x and u:

$$J(x,u) = \frac{1}{2\pi}\int\limits_{-\infty}^{+\infty} \frac{1}{2}\bigl(X(\omega)^{\mathrm{H}}LX(\omega) + U(\omega)^{\mathrm{H}}MU(\omega)\bigr)\,d\omega$$

where X and U satisfy (5.1).

Thus, this optimal control problem is reduced to the problem of minimization of the functional

$$\int\limits_{-\infty}^{+\infty} \Bigl[\bigl(G_0(i\omega)x_0 + G_u(i\omega)U(\omega)\bigr)^{\mathrm{H}}L\bigl(G_0(i\omega)x_0 + G_u(i\omega)U(\omega)\bigr) + U(\omega)^{\mathrm{H}}MU(\omega)\Bigr]\,d\omega$$

over the set of all admissible U.

Note that by completing the square this quadratic functional can be represented in the form

$$\int\limits_{-\infty}^{+\infty} \Bigl(x_0^{\mathrm{T}}Q(i\omega)x_0 + \bigl\|S^{-1/2}(i\omega)U(\omega) + S^{1/2}(i\omega)G_u^{\mathrm{H}}(i\omega)LG_0(i\omega)x_0\bigr\|^2\Bigr)\,d\omega \tag{5.5}$$

where

$$Q(i\omega) := G_0^{\mathrm{H}}(i\omega)LG_0(i\omega) - G_0^{\mathrm{H}}(i\omega)LG_u(i\omega)S(i\omega)G_u^{\mathrm{H}}(i\omega)LG_0(i\omega). \tag{5.6}$$

It is obvious that the minimum of the functional J (1.4) is attained at the optimal control u whose Fourier transform satisfies

$$U(\omega) = -S(i\omega)G_u^{\mathrm{H}}(i\omega)LG_0(i\omega)x_0. \tag{5.7}$$

Since the optimal control (as it follows from the minimum principle stated in Theorem 2.1) is continuous on $[0,+\infty)$, we can find $u(0)$ by using the inverse Fourier transform:

$$\frac{1}{2}u(0) = \frac{1}{2\pi}\int\limits_{-\infty}^{+\infty} U(\omega)\,d\omega. \tag{5.8}$$

Because of (5.7) we obtain the following expression for $u(0)$:

$$u(0) = -\frac{1}{\pi}\left(\int_{-\infty}^{+\infty} S(i\omega)G_u^{\mathrm{H}}(i\omega)LG_0(i\omega)\,d\omega\right)x_0.$$

This implies that the optimal feedback satisfies (5.3).

We also find from (5.5) that the optimal value of the linear–quadratic functional J (1.4) coincides with

$$\frac{1}{4\pi}\int_{-\infty}^{+\infty} x_0^{\mathrm{H}}Q(i\omega)x_0\,d\omega.$$

Because of (3.1) (where R is a solution of the matrix Riccati equation) we obtain the representation (5.4). The theorem is proven. $\square$

CONCLUSION

In this paper we suggested two methods for the derivation of analytical formulas for solutions of matrix Riccati equations related to linear–quadratic optimal control problems. The first method is based on the necessary optimality conditions stated in terms of the linear adjoint system of equations and exploits elementary properties of the Laplace transform.

We also demonstrated a useful formalism for finding analytical solutions of more general algebraic and differential matrix quadratic equations which is based on similar linear systems. It is possible to consider an analogous approach to infinite-dimensional systems with unbounded operators.

Another interesting direction for research could be the development of analogous methods for control systems on manifolds.

The second method is based on the reduction of the original optimal control problem to a problem of quadratic optimization in the space of Fourier transforms of controls. Its advantage is the representation formula for the optimal stabilizing feedback only in terms of matrix transfer functions.

We should note that this method can also be used for the analysis of minimax problems related to the H^∞ robust control approach. These results will appear in a forthcoming paper.

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